

Wide Bandgap Device/PI Engineer (R&D)

Texas Instruments · Dallas, TX, United States

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Description

Change the world. Love your job.

TI is seeking an innovative and highly skilled GaN Device Project Lead Engineer to join our cutting-edge Gallium Nitride (GaN) Technology Development Team. This role offers the unique opportunity to work at the forefront of developing Gallium Nitride technology that is redefining the power electronics landscape — enabling higher efficiency, faster switching, and smaller, more reliable systems. You will be instrumental in developing and scaling GaN power process technologies for high-volume manufacturing. You'll collaborate across disciplines to deliver state-of-the-art devices with world-class performance and reliability.

About the job

This position is focused on leading the development, characterization and optimization of gallium nitride (GaN) power devices. As part of TI's Advanced Technology Development (ATD) team, the role involves close collaboration with fab/manufacturing, business units and Kilby Labs. This position requires in-depth knowledge of III-nitride materials, device physics, and reliability. The scope of the job includes, but not limited to:

- GaN device simulations (TCAD), device design, layout

- Develop robust process flows, drive design-of-experiment (DOE) execution, and solve complex integration challenges
- Collaboration with fab/manufacturing engineers to ensure device manufacturability
- Collaboration with reliability engineers to meet device reliability requirements
- Failure root-cause investigation for device/process/reliability/yield improvement

The ideal candidate is expected to have a strong understanding of the following:

- Power FETs and other relevant semiconductor device physics
- Typical analog silicon and GaN high electron mobility transistor (HEMT) process flows and the underlying process physics
- Electrical and materials characterization methodologies and statistical data analysis
- Strong hands-on lab experience and excellent problem solving skills are also needed
- Excellent verbal and written communication skills are a must as the position requires interfacing with multiple teams

Qualifications

Minimum requirements:

- Master's in Electrical Engineering, Electrical & Computer Engineering, Materials Engineering, Physics, or related degree
- 3+ years of hands-on experience in GaN or compound semiconductor device/process development
- Strong knowledge of semiconductor processing, especially in III-V materials
- Experience with process integration, DOE planning, electrical characterization, and failure analysis
- Proven ability to work across functional teams and drive results in a fast-paced R&D environment

Preferred qualifications:

- PhD in Electrical Engineering, Electrical & Computer Engineering, Materials Engineering, Physics, or related degree
- Familiarity with GaN-on-Silicon technology for power switching applications
- Experience with fab tools and process modules/Integration
- Strong communication and data analysis skills
- Knowledge of device physics, reliability mechanisms, and packaging interaction is a plus.
- Ability to work in teams and collaborate effectively with people in different functions
- Ability to take the initiative and drive for results
- Sound decision-making capabilities and the ability to prioritize tasks and adapt accordingly

About the organization

TI does not make recruiting or hiring decisions based on citizenship, immigration status or national origin. However, if TI determines that information access or export control restrictions based upon applicable laws and regulations would prohibit you from working in this position

without first obtaining an export license, TI expressly reserves the right not to seek such a license for you and either offer you a different position that does not require an export license or decline to move forward with your employment.

About the company

Why TI?

- Engineer your future. We empower our employees to truly own their career and development. Come collaborate with some of the smartest people in the world to shape the future of electronics.
- We're different by design. Diverse backgrounds and perspectives are what push innovation forward and what make TI stronger. We value each and every voice, and look forward to hearing yours. [Meet the people of TI](#)
- Benefits that benefit you. We offer competitive pay and benefits designed to help you and your family live your best life. Your well-being is important to us. Please find our country-specific benefits [here](#)

About Texas Instruments

Texas Instruments Incorporated (Nasdaq: TXN) is a global semiconductor company that designs, manufactures and sells analog and embedded processing chips for markets such as industrial, automotive, data center, personal electronics and communications equipment. At our core, we have a passion to create a better world by making electronics more affordable through semiconductors. This passion is alive today as each generation of innovation builds upon the last to make our technology more reliable, more affordable and lower power, making it possible for semiconductors to go into electronics everywhere. Learn more at [TI.com](https://www.ti.com).

Texas Instruments is an equal opportunity employer and supports a diverse, inclusive work environment. All qualified applicants will receive consideration for employment without regard to race, color, religion, creed, disability, genetic information, national origin, gender, gender identity and expression, age, sexual orientation, marital status, veteran status, or any other characteristic protected by federal, state, or local laws.

If you are interested in this position, please apply to this requisition.